

This controls the overall polygon count of the animated scene that the computer must render in real time.

Deforming/modifying

Deformation is crucial for modeling. It's a great way to mold your object

in a way that would be difficult polygon by polygon. Moving a vertex, an edge, or a face is the most basic kind of deforming. Popular 3D programs allow use of brushes to modify the models. They come in different shapes and sizes and can be used to smooth, dodge, inflate and pinch.



Subdividing

Creating a rigid body

There are also surfaces that you'll not want to be deformed during dynamic animations. In those cases, you'll want

to make the surface a rigid body. Rigid bodies are great for things such as robots, cars, and wooden swords etc which do not change their course during the game.



Polygon count

Texturing: Texturing/texture mapping is the technique of applying textures/color to the 3D models. Textures are used to give real-world look to 3D models and are created using tools such as Adobe Photoshop, Illustrator etc. One polygon can have more than one texture. This is called multi-texturing and is used to create complex effects.

Preparing 3D model for animators

In games, these 3D models you make need to look realistic and thus need to be animated. Thus it is important to prepare your model for animation. Some basic concepts that are required in this process include:

- ▶ **Creating soft bodies:** Ever thought how will you model things like a flag or hair etc which react to forces like wind, movement etc. These objects need a different type of modeling than polygonal meshes. They need to be converted into soft-bodies. When you make your model a soft body, you copy the vertices in the mesh and replace the copied vertices with particles. Particles are virtual points that react to simulated forces or even collisions with other objects. Movement of these particles is dynamic and their animation is controlled by forces that act on these particles making it more realistic as compared to manual animation.